BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 055 · Lesson 055 · Choosing Data Structures for Problems

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a hash collision?

Q2. When can linked-list insertion be O(1)?

Q3. What is a BST's advantage when balanced?

Q4. 021

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

operation — Action such as lookup or insertion.

random access — Direct access by numeric index.

trade-off — A benefit weighed against a cost.

complexity — Growth in resource use.

New learning

Discover — Choose structures from operations first, not familiar names: indexed access, insertion, deletion, queue discipline or key lookup.

Trace — Consider runtime and additional memory: arrays/lists support indexed access; linked lists can insert cheaply given a node; dictionaries use extra bucket storage.

Evaluate — Justify the suitability for a real task, including constraints and the cost of maintaining the structure.

Worked example

Choose a structure for a browser's Back action and justify.

1. Explain the trace — Stack (LIFO): the most recently visited earlier page is retrieved first. Browser navigation may use multiple stacks for back/forward.

We Do · Practise with guidance

Choose a structure for clinic tickets processed in arrival order.

```
queue.Enqueue(ticket)
next=queue.Dequeue()
```

Explain the trace

Queue FIFO serves earliest arrivals first, subject to real-world priority policies.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Match an undo feature, printer queue, pupil ID lookup and road map to a structure. [4]

Question 2

A timetable stores fixed grid of rooms and periods. Recommend a 2D array or graph and explain operations. [6]

Question 3

Evaluate using List<T> versus linked list for a million-item collection, focusing on frequent random index access versus middle insertion with a known node. [6]

Exit Ticket · Answer before checking

The same data structure is always optimal for every collection.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Two different keys map to the same bucket.

Q2. When a reference to the insertion node is already known.

Q3. Ordered search can require O(log n) comparisons.

Q4. Push 4,8,2 into stack then Pop, versus enqueue 4,8,2 then Dequeue. Give outputs.

You Do pages · Independent questions

Q1. Stack [1], Queue [1], Dictionary/hash table [1], Graph [1].

Q2. 2D array room×period [2]; indexed direct access [2]; fixed rectangular dimensions suit [1]; graph unnecessary for this access [1].

Q3. List random indexing O(1) [2]; LinkedList random index traversal O(n) [1]; List middle insertion O(n) shifts [1]; linked list pointer insertion with known node O(1) [2].

Exit Ticket · Model answer

False. Choose from operations, sizes and constraints.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.